

EDUCATION

- **Master of Science in Data Science** Berlin, Germany
Berlin University of Applied Sciences (BHT) — GPA: 3.92 Oct. 2024 – Sept. 2026
- **Bachelor of Science in Computer Science; Minors in Math & Stats** Bellingham, WA
Western Washington University — GPA: 3.96 Sept. 2020 – June 2024

EXPERIENCE

- **Data Science Researcher (Machine Learning & Engineering)** Berlin, Germany
Calgo Lab — Berlin University of Applied Sciences (BHT) Feb. 2025 – Sept. 2026
 - Engineered distributed ML experiment pipelines on a Kubernetes cluster with Helm; automated testing and linting via GitHub Actions CI/CD.
 - Deployed and maintained Dockerized LLM serving stack (vLLM on Nvidia GPUs) and production web-demo applications.
 - Architected **MechDetect**, a novel ML framework for classifying error mechanisms in tabular datasets; published at IEEE DSIS 2025.
 - Implemented spatiotemporal groundwater-level forecasting (master's thesis) combining Chronos2, N-HiTS, and GPyTorch Gaussian Processes (Kronecker product, Matérn) across 499 monitoring sites.
- **Computer Science Researcher (Deep Learning)** Bellingham, WA
Hutch Research — Western Washington University Mar. 2023 – June 2024
 - Built a PyTorch/TensorFlow probabilistic neural network pipeline to predict binary star system parameters; published in *The Astronomical Journal*, Vol. 171 (2026).
 - Optimized large astronomical dataset processing with NumPy; maintained code quality and reproducibility in a 4-person team.
- **Mathematics Researcher (Statistical Analysis)** Bellingham, WA
Western Washington University Oct. 2022 – June 2024
 - Developed a novel class of non-parametric two-sample tests in R; presented at Joint Mathematics Meetings 2024 and SIAM PNW 2023.

SELECTED PROJECTS

- **LangGraph Email Agent** | *vLLM, LangGraph, Kubernetes, Docker, Nvidia A100*
 - Deployed an agentic LLM email-answering system on Kubernetes using vLLM for inference serving and LangGraph for multi-step orchestration with structured output.
- **Tax Law RAG System** | *BERT Fine-tuning, LangChain, FAISS, Kubernetes*
 - Built a production RAG pipeline with fine-tuned BERT retrieval; benchmarked FAISS vs. Annoy vector databases on English and German corpora. Published the fine-tuned BERT to HuggingFace.
- **Berlin PM2.5 Spatiotemporal Modeling** | *Chronos2, Kubernetes, Docker, Streamlit*
 - Automated daily ETL and inference pipeline on Kubernetes; spatiotemporal forecasting via Chronos2 foundation model and ordinary kriging.
- **tab-err: Tabular Data Error Library** | *Python, PyPI, CI/CD, GitHub Actions*
 - Contributed to open-source library for realistic tabular error generation; developed high-level APIs and comprehensive unit tests.

PUBLICATIONS

- **MechDetect: Detecting Data-Dependent Errors** | IEEE DSIS 2025 DOI: 10.1109/DSIS67228.2025.11390600 2026
 - Architected and experimentally evaluated a novel ML framework for characterizing error mechanisms in tabular datasets.
- **Towards Realistic Error Models for Tabular Data** | ACM JDIQ DOI: 10.1145/3774914 2025
 - Integrated tab-err into a Kubernetes/Helm framework to benchmark scalability and downstream ML impact of tabular errors.
- **Probabilistic NNs for Eclipsing Binary Parameters** | *The Astronomical Journal* 2026
 - Prototyped and evaluated neural network architectures on large astronomical datasets; published in Vol. 171, No. 5. DOI: 10.3847/1538-3881/ae5b79

SKILLS

MLOps & Deployment: Kubernetes, Docker, Helm, vLLM, TorchServe, FastAPI, GitHub Actions, Prometheus, Grafana, REST/gRPC API design
LLM & Agentic AI: LangChain, LangGraph, FAISS, RAG, Prompt Engineering, Structured Outputs, Agentic Pipelines, Transformer Fine-tuning
ML & Data: PyTorch, TensorFlow, GPyTorch, Scikit-learn, Pandas, Polars, NumPy, PostgreSQL, MySQL, Time-Series, Spatial Data, IML/XAI, Wandb
Cloud & DevOps: AWS, GPU Workloads (Nvidia A100), Containerization, Agile / Git-based workflows, Linux
Languages: Python, SQL, R, Go, Bash, Java **Spoken:** English (Native), German (B1)